Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18SE7DE43]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SECURE PROGRAMMING

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1.	a.	List out the ten principles for building secure software while developing software and assisting in the prevention of frequent security issues.	07
	b.	Race conditions are a common occurrence in multithreaded programming, whether or not the term is used. How the race condition results in vulnerability of the system.	07
2.	a.	Database security measures safeguard the database management system and all applications that access it from infiltration, data misuse, and damage. Elaborate the process of protecting the database from intruders.	07
	b.	The workers click the links, surf the target website for a period of time, and possibly sign up for newsletters prior to clicking another link. Categorize the methodologies in creating the fake ads and accessing websites automatically and obtaining data.	07
3.	a.	Intruder exploits the program when more data is put into a fixed-length buffer than the buffer can handle. Summarize the attack with an example	07
	b.	The Format String exploit occurs when the submitted data of an input string is evaluated as a command by the application. Explain about the Format string vulnerability.	07

OR

4.	a.	Software security is the idea of engineering software so that it continues to function correctly under malicious attack. Discuss about the software security fundamentals.	07
	b.	Determine how the web security vulnerability that allows an attacker to interfere with the queries that an application makes to its database.	07
5.	a.	Identify the attack which results in the linking of an attacker's MAC address with the IP address of a legitimate computer or server on the network. Discuss in detail.	07
	b.	The attack is referred to as DNS redirection, the process is utilized by hackers to alter the resolution of a Domain Name System (DNS), using malware that ensures the authentic server is modified to not comply with the set internet standards. Identify the counter measures of the attack.	07

OR

6. a. Identify the impact of an application, if the improper use of cryptography. **07**

b. Analyze which protocol can be used for establishing a secure channel between two devices that are connected over the Internet or an internal connection. Describe in detail. **07**

7. a. Rogue DHCP Server Attack in an addition to DHCP starvation attacks, when attacker can perform MITM attacks such as sniffing, An attacker who succeeds in exhausting the DHCP Server's IP address space can found out a Rogue DHCP Server on the network which isn't under the control of the network administrator. Explain the vulernabilty cause by Rogue DHCP Server Attack. **07**

b. If you're developing custom software and applications, you have to build it on a foundation of robust security. However, for a variety of reasons, enterprises across industries continue to make cyber security a low priority. Discuss the importance of role played by Software Tester. **07**

OR

8. a. Web application testing, a software testing technique exclusively adopted to test the applications that are hosted on web in which the application interfaces and other functionalities are tested. Discuss the Web testing in detail. **07**

b. Many sites implement simple restrictions based on a list of permitted (or blocked) extensions, this is not sufficient to prevent attackers from uploading legitimate file types that have malicious contents. What kind of testing is required for ensuring the security of file? **07**